

Description:

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit**1. Download Sleuth Kit:**

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

1. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

3. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

3. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

css

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

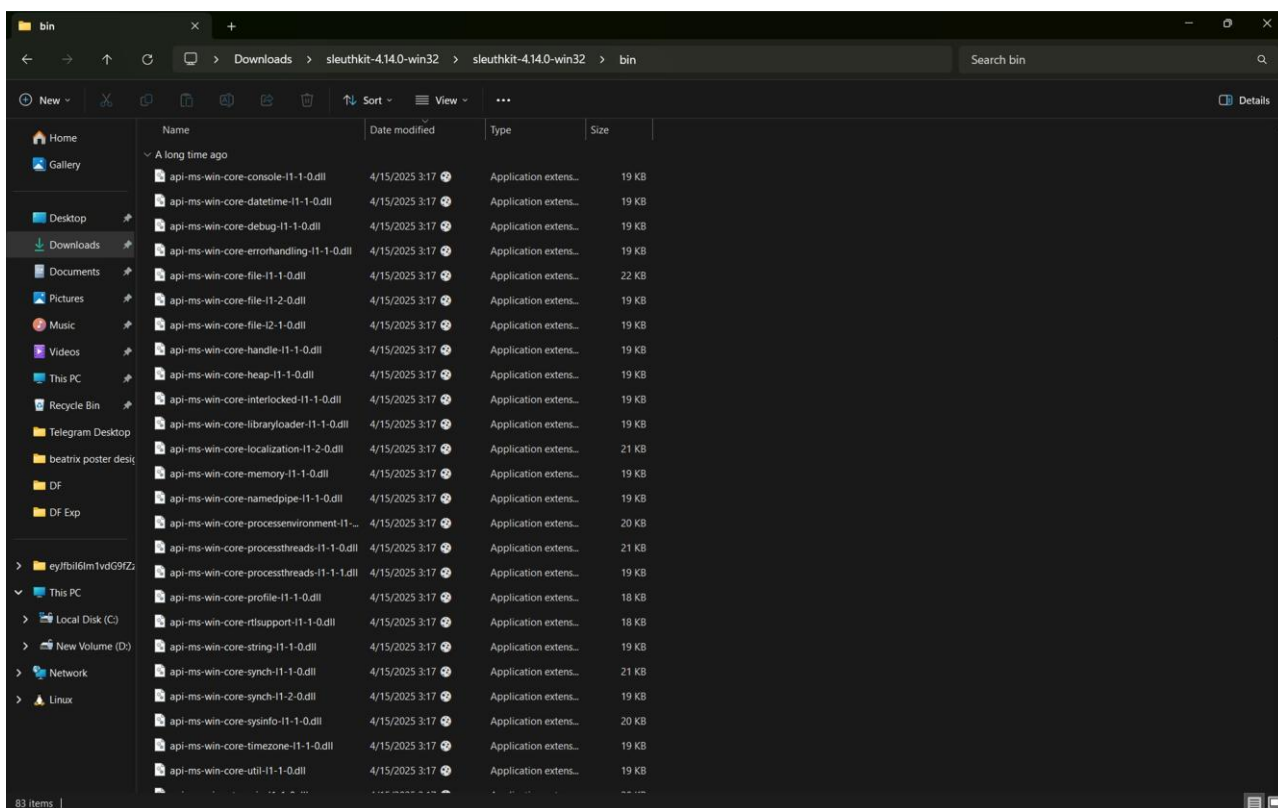
- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation

Fsstat command execution:



Identifying File System:

```
C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
Missing image name
Usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] i
  -t: display type only
  -i imgtype: The format of the image file (use '-i list' for supported types)
  -b dev_sector_size: The size (in bytes) of the device sectors
  -f fstype: File system type (use '-f list' for supported types)
  -o imgoffset: The offset of the file system in the image (in sectors)
  -P pooltype: Pool container type (use '-P list' for supported types)
  -B pool_volume_block: Starting block (for pool volumes only)
  -v: verbose output to stderr
  -V: Print version
  -k password: Decryption password for encrypted volumes

C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

Installed Sleuth Kit:

```

C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -
de CPi (1).E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)   Size: 48-72   Flags: Resident
$ATTRIBUTE_LIST (32)        Size: No Limit  Flags: Non-resident
$FILE_NAME (48)             Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)             Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)    Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)           Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)    Size: 12-12   Flags: Resident
$DATA (128)                 Size: No Limit  Flags:

```

Identifying File System:

```

C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-
win32\bin>fls -i ewf -o 63 -r "C:\Users\chenna\Downloads\sleuthkit-4.14.0-
win32\4Dell Latitude CPi (1).E01" > file_list.txt

```

Recovering Deleted File:

```

-/d 3671-144-7: Documents and Settings
+ -/d 344-144-6:      chenna
++ -/r 345-128-4:     NTUSER.DAT
++ -/d 346-144-6:     Templates
+++ r/r 9760-128-4:    amipro.sam
+++ r/r 9759-128-4:    excel.xls
+++ r/r 9758-128-4:    excel4.xls
+++ r/r 9757-128-4:    lotus.wk4
+++ r/r 9756-128-4:    powerpnt.ppt
+++ r/r 9755-128-3:    presenta.shw
+++ r/r 9754-128-4:    quattro.wb2
+++ r/r 9753-128-3:    sndrec.wav
+++ r/r 9752-128-4:    winword.doc
+++ r/r 9750-128-4:    winword2.doc
+++ r/r 9751-128-3:    wordpfct.wpd
+++ r/r 9749-128-3:    wordpfct.wpg
++ -/d 3665-144-1:     Start Menu
+++ r/r 9778-128-3:    desktop.ini
+++ d/d 4156-144-5:    Programs
++++ -/d 4186-144-1:    Startup
+++++ r/r 9762-128-3:    desktop.ini
+++++ -/d 4746-144-6:    Accessories
+++++ -/d 5502-144-1:    Entertainment
+++++ r/r 9770-128-3:    desktop.ini

```

Recovering Deleted File:

```
C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ewf -o 63 "C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" 150 > recovered.txt
```

```
C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

REGEDIT4

```
[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000507-0000-0010-8000-00AA006D2EA4}\InprocServer32]  
"ThreadingModel"="Both"
```

```
[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000514-0000-0010-8000-00AA006D2EA4}\InprocServer32]  
"ThreadingModel"="Both"
```

```
[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{0000050B-0000-0010-8000-00AA006D2EA4}\InprocServer32]  
"ThreadingModel"="Both"
```

```
[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000535-0000-0010-8000-00AA006D2EA4}\InprocServer32]  
"ThreadingModel"="Both"
```

```
[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000541-0000-0010-8000-00AA006D2EA4}\InprocServer32]  
"ThreadingModel"="Both"
```

Metadata:

```
C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i  
e CPi (1).E01" 150
```

MFT Entry Header Values:

```
Entry: 150          Sequence: 150  
$LogFile Sequence Number: 9239131  
Allocated File  
Links: 1
```

\$STANDARD_INFORMATION Attribute Values:

```
Flags: Archive  
Owner ID: 0  
Security ID: 0 ()  
Version 16 of 16943392  
Created:      1999-04-24 03:52:00.000000000 (India Standard Time)  
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)  
MFT Modified:  2004-08-19 22:32:11.098512000 (India Standard Time)  
Accessed:      2004-08-18 05:30:00.000000000 (India Standard Time)
```

\$FILE_NAME Attribute Values:

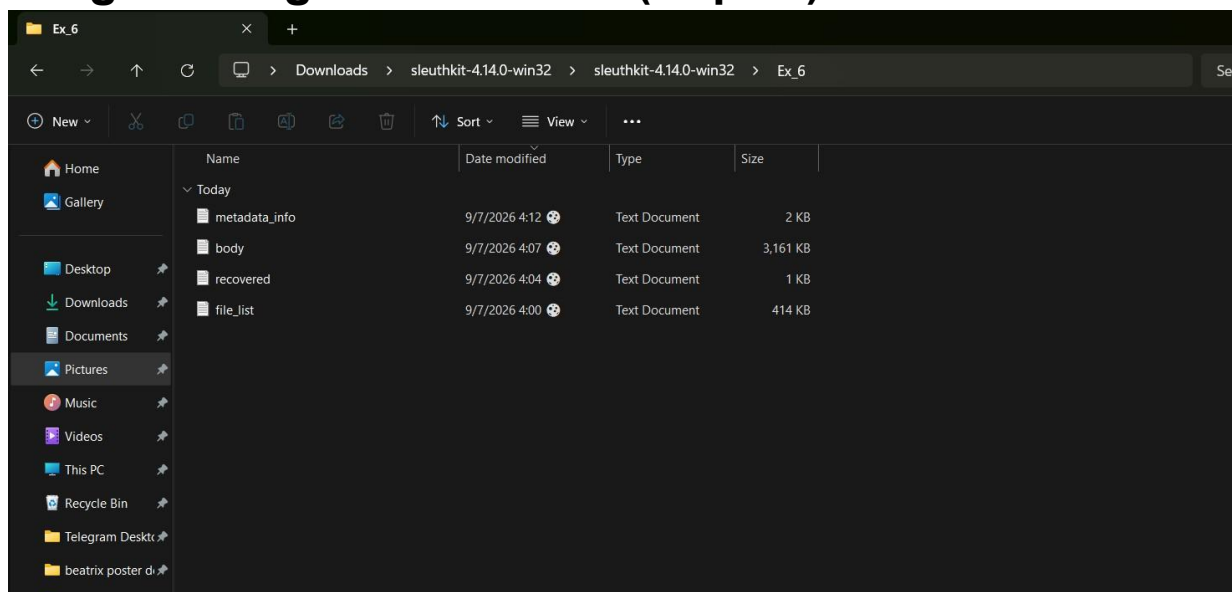
```
Flags: Archive  
Name: ADOFRE15.REG  
Parent MFT Entry: 137   Sequence: 137  
Allocated Size: 1024   Actual Size: 640  
Created:      1999-04-24 03:52:00.000000000 (India Standard Time)  
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)  
MFT Modified:  1999-04-24 03:52:00.000000000 (India Standard Time)
```

Generating Timeline:

```
C:\Users\chenna\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-  
win32\bin>fls -i ewf -o 63 -m / -r "C:\Users\chenna\Downloads\sleuthkit-  
4.14.0-win32\4Dell Latitude CPi (1).E01" > body.txt
```

```
0|/Documents and Settings ($FILE_NAME)|3671-48-2|-/drwxrwxrwx|0|0|110|1092934741|1092934741|1092934741|1092934741  
0|/Documents and Settings|3671-144-7|-/drwxrwxrwx|0|0|56|1093619285|1092956645|1092956645|1092934741  
0|/Documents and Settings/chenna ($FILE_NAME)|344-48-2|-/drwxrwxrwx|0|0|82|1092956645|1092956645|1092956645|1092956645  
0|/Documents and Settings/chenna|344-144-6|-/drwxrwxrwx|0|0|56|1093621260|1093621260|1093621260|1092956645  
0|/Documents and Settings/chenna/NTUSER.DAT ($FILE_NAME)|345-48-5|-/rr-xr-xr-x|0|0|86|1092956645|1092955127|1092956645  
0|/Documents and Settings/chenna/NTUSER.DAT|345-128-4|-/rr-xr-xr-x|0|0|786432|1093621583|1093621583|1093621573|1092956645  
0|/Documents and Settings/chenna/Templates ($FILE_NAME)|346-48-2|-/dr-xr-xr-x|0|0|84|1092956645|1092956645|1092956645|1092956645  
0|/Documents and Settings/chenna/Templates|346-144-6|-/dr-xr-xr-x|0|0|56|1093015079|1092954275|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/amipro.sam ($FILE_NAME)|9760-48-5|r/rrwxrwxrwx|0|0|86|1092956646|998589600|1092954273|1092956646  
0|/Documents and Settings/chenna/Templates/amipro.sam|9760-128-4|r/rrwxrwxrwx|0|0|4570|1092956646|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/excel.xls ($FILE_NAME)|9759-48-5|r/rrwxrwxrwx|0|0|84|1092956646|998589600|1092954273|1092956645  
0|/Documents and Settings/chenna/Templates/excel.xls|9759-128-4|r/rrwxrwxrwx|0|0|5632|1092956646|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/excel4.xls ($FILE_NAME)|9758-48-5|r/rrwxrwxrwx|0|0|84|1092956645|998589600|1092954273|1092956645  
0|/Documents and Settings/chenna/Templates/excel4.xls|9758-128-4|r/rrwxrwxrwx|0|0|1518|1092956645|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/lotus.wk4 ($FILE_NAME)|9757-48-5|r/rrwxrwxrwx|0|0|84|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/lotus.wk4|9757-128-4|r/rrwxrwxrwx|0|0|2448|1092956645|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/powerpnt.ppt ($FILE_NAME)|9756-48-5|r/rrwxrwxrwx|0|0|90|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/powerpnt.ppt|9756-128-4|r/rrwxrwxrwx|0|0|12288|1092956645|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/presenta.shw ($FILE_NAME)|9755-48-4|r/rrwxrwxrwx|0|0|90|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/presenta.shw|9755-128-3|r/rrwxrwxrwx|0|0|461|1092956645|998589600|1092956645|1092956645  
0|/Documents and Settings/chenna/Templates/quattro.wb2 ($FILE_NAME)|9754-48-5|r/rrwxrwxrwx|0|0|88|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/quattro.wb2|9754-128-4|r/rrwxrwxrwx|0|0|4017|1092956645|998589600|1092956645|1092956645  
0|/Documents and Settings/chenna/Templates/sndrec.wav ($FILE_NAME)|9753-48-4|r/rrwxrwxrwx|0|0|86|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/sndrec.wav|9753-128-3|r/rrwxrwxrwx|0|0|58|1092956645|998589600|1092956645|1092956645  
0|/Documents and Settings/chenna/Templates/winword.doc ($FILE_NAME)|9752-48-5|r/rrwxrwxrwx|0|0|88|1092956645|998589600|1092954274|1092956645  
0|/Documents and Settings/chenna/Templates/winword.doc|9752-128-4|r/rrwxrwxrwx|0|0|4608|1092956645|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/winword2.doc ($FILE_NAME)|9750-48-5|r/rrwxrwxrwx|0|0|90|1092956645|998589600|1092954275|1092956645  
0|/Documents and Settings/chenna/Templates/winword2.doc|9750-128-4|r/rrwxrwxrwx|0|0|1769|1092956645|998589600|1092956646|1092956645  
0|/Documents and Settings/chenna/Templates/wordpfct.wpd ($FILE_NAME)|9751-48-4|r/r-wx-wx-wx|0|0|90|1092956645|998589600|1092954275|1092956645  
0|/Documents and Settings/chenna/Templates/wordpfct.wpd|9751-128-3|r/r-wx-wx-wx|0|0|30|1092956645|998589600|1092956645|1092956645  
0|/Documents and Settings/chenna/Templates/wordpfct.wpg ($FILE_NAME)|9749-48-4|r/r-wx-wx-wx|0|0|90|1092956645|998589600|1092954275|1092956645  
0|/Documents and Settings/chenna/Templates/wordpfct.wpg|9749-128-3|r/r-wx-wx-wx|0|0|57|1092956645|998589600|1092956645|1092956645  
0|/Documents and Settings/chenna/Start Menu ($FILE_NAME)|3665-48-2|-/d-wx-wx-wx|0|0|86|1092956645|1092956645|1092956645|1092956645  
0|/Documents and Settings/chenna/Start Menu|3665-144-1|-/d-wx-wx-wx|0|0|256|1093619286|1092934809|1092956646|1092956645
```

Gathering all the generated Files(Report):



Rubrics

Criteria&Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity&Submission Regularity	3	
2. Application of Forensic Tools&Practical Execution	3	
3. Documentation&Reporting	2	
4. Engagement, Problem-Solving&Team Collaboration	2	
Total	10	

Result: The experiment successfully analyzed the digital evidence image and extracted crucial forensic artifacts using The Sleuth Kit (TSK) command-line utilities. Using fls and icat, the investigator recursively mapped the directory structure and **recovered deleted files via their inode numbers**, while mactime successfully generated a complete chronological timeline of system activity based on MAC timestamps.

